Smart Dine API Documentation

This document provides detailed information about the Smart Dine REST API endpoints.

Base URL

```
http://localhost:8000/api
```

Authentication

All API requests require authentication except for public customer endpoints. Include the JWT token in the Authorization header:

```
Authorization: Bearer <your-jwt-token>
```

Authentication Endpoints

POST /auth/login

Login with email and password.

Request Body:

```
{
  "email": "admin@smartdine.local",
  "password": "Admin@12345"
}
```

Response:

```
{
  "access_token": "eyJ...",
  "token_type": "bearer",
  "user": {
    "id": "uuid",
    "email": "admin@smartdine.local",
    "full_name": "Admin User",
    "role": "ADMIN",
    "branch_id": 1
  }
}
```

POST /auth/register

Register a new user (admin only).

GET /auth/me

Get current user information.

Core API Endpoints

Branches

- `GET /branches` - List all branches
- `POST /branches` - Create new branch
- `GET /branches/{id}` - Get branch details
- `PUT /branches/{id}` - Update branch
- `DELETE /branches/{id}` - Delete branch

Tables

- `GET /tables` - List dining tables
- `POST /tables` - Create new table
- `GET /tables/{id}` - Get table details
- `PUT /tables/{id}` - Update table (position, status, etc.)
- `DELETE /tables/{id}` - Delete table

Menu Management

- `GET /menu/categories` - List menu categories
- `POST /menu/categories` - Create category
- `GET /menu/items` - List menu items
- `POST /menu/items` - Create menu item
- `PUT /menu/items/{id}` - Update menu item
- `DELETE /menu/items/{id}` - Delete menu item

Orders

- `GET /orders` - List orders (filtered by status, date, etc.)
- `POST /orders` - Create new order
- `GET /orders/{id}` - Get order details
- `PUT /orders/{id}` - Update order (status, items, etc.)
- `PUT /orders/{id}/status` - Update order status
- `DELETE /orders/{id}` - Cancel order

Kitchen

- `GET /kitchen/orders` - Get orders for kitchen (pending/preparing)
- `PUT /kitchen/orders/{order_id}/items/{item_id}/status` - Update item preparation status

Billing

- `GET /billing/orders/{order_id}` - Get order for billing
- `POST /billing/orders/{order_id}/payment` - Process payment
- `GET /billing/invoice/{order_id}` - Generate PDF invoice

Inventory

- `GET /inventory` - List inventory items
- `POST /inventory` - Add inventory item
- `PUT /inventory/{id}` - Update inventory
- `DELETE /inventory/{id}` - Remove inventory item
- `GET /inventory/low-stock` - Get low stock alerts

Suppliers

- `GET /suppliers` - List suppliers
- `POST /suppliers` - Add supplier
- `PUT /suppliers/{id}` - Update supplier
- `DELETE /suppliers/{id}` - Remove supplier

Employees

- `GET /employees` - List employees
- `POST /employees` - Add employee
- `PUT /employees/{id}` - Update employee
- `DELETE /employees/{id}` - Remove employee
- `POST /employees/{id}/attendance` - Record attendance

Reports

- `GET /reports/sales` - Sales reports
- `GET /reports/inventory` - Inventory reports
- `GET /reports/employee` - Employee reports

Insights

- `GET /insights/recommendations` - Get business recommendations
- `GET /insights/sales-prediction` - Sales predictions

Public Customer API

These endpoints don't require authentication and are used by the customer-facing app.

GET /public-customer/menu

Get public menu for a specific table via QR token.

Query Parameters:

- `token` - QR token from table

POST /public-customer/orders

Place an order from customer interface.

Request Body:

```
{
  "table_qr_token": "abc123",
  "items": [
    {
      "menu_item_id": 1,
      "quantity": 2,
      "special_instructions": "No onions"
    }
  ],
  "customer_name": "John Doe"
}
```

WebSocket API

Connection

```
ws://localhost:8000/ws/orders
```

Messages

Subscribe to order updates:

```
{
  "type": "subscribe",
  "order_ids": [1, 2, 3]
}
```

Order update broadcast:

```
{
  "event": "order_updated",
  "order_id": 1,
  "status": "preparing",
  "updated_items": [...]
}
```

Error Responses

All endpoints return standard HTTP status codes. Error responses include:

```
{
  "detail": "Error message"
}
```

Common status codes:

- `200` - Success
- `201` - Created
- `400` - Bad Request
- `401` - Unauthorized
- `403` - Forbidden
- `404` - Not Found
- `422` - Validation Error
- `500` - Internal Server Error

Rate Limiting

API endpoints are rate-limited to prevent abuse. Contact admin for limits.

File Uploads

Menu item images can be uploaded to `/uploads/menu/`. Use multipart/form-data.

Pagination

List endpoints support pagination:

- `page` - Page number (default: 1)
- `limit` - Items per page (default: 50)

Response includes:

```
{
  "items": [...],
  "total": 100,
  "page": 1,
  "pages": 2
}
```